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Articles Appreciated

The January issue of EFY is very informative and useful. Though all articles are well written and useful, some articles like 'The Semiconductors and World's Energy Crisis' by Vivek Sharma, 'Ensuring LEDs' Long Life' by Reinhard Zimmermann and 'Latest in Chip Fabrication' by Uma Gupta need special mention.

In Vivek's article, I would like to add that variable-frequency AC motor drives, which use integral motors designed for electric vehicles, may have good use in energy-efficient electric vehicles. Replacement of silicon in photovoltaic devices for solar energy does not appear to be a near-term possibility.

Congratulations to EFY for bringing out such a useful publication, which should be taken seriously by government policy makers and implementation agencies in both private and public sector.

Dr G.N. Acharya
Pune

☐ I am an electronics and communications engineering graduate willing to study more and work in the embedded systems space. The 'Embedded Systems: Fast Track For Smart Brains' article by Uma Gupta has been really helpful to me. Thank you!

Construction and Circuit Ideas Listing

I have been reading EFY for the last 20 years. I keep separate records of the published circuit ideas and construction projects. But recently I could not maintain the record due to paucity of time. I request you to publish the topic-wise list of projects published in EFY in the last 20 years or from the time of its first publication. There should be separate lists for circuit ideas, constructions and software projects with month and year of publication.

Uttam Dutta
Through e-mail

Solar Charger

Thanks for publishing 'Photovoltaic Battery Charge Controller' in January 2011 issue. I have assembled only the

charger section and used it for a 400-watt solar panel. The charger is working satisfactorily. I want to modify this charger to charge two 12V, 150Ah batteries or a 24V battery. Please suggest how to do it.

A.D. Kamble
Through e-mail

The author Gp Capt. (ret'd) K.C. Bhaskin replies: I am pleased to note that the reader has benefited from the published construction project on the subject.

For the first part of the query, I presume that his 400-watt solar panel comprises four individual solar panels rated at 18V, 125-watt each, connected in parallel to deliver 400-watt power for charging 12V batteries. With this configuration, one may charge up to three batteries of 12V, 150Ah capacity or even four 12V, 100Ah batteries.

As regards the second part of the query, it is not clear whether the load is meant for 12V or 24V. However, my charger circuit is designed for use with 12V, 80Ah batteries. It could also work for 12V, 100Ah batteries. But if one wants to charge two 12V, 150Ah batteries (connected in parallel only), the electric charge capacity becomes 300Ah (at 12V). Hence your circuit's charging current (as well as load current at 12V) would be about three times higher. The circuit needs to be modified (refer Fig. 7 on page 108 in the article) to ensure that the safe current rating of components (such as diode D1, MOSFETs T1 and T2 and fuse F1) is about 30A and the value of sampling resistors (R5 and R6) is 0.033 ohm each, with a power rating of 15 to 20 watts.

For a fully satisfactory reply, please forward the technical specifications of the panels used (including maximum power voltage and maximum power current) and the configuration (number of panels used and series/parallel combination in case more than one panel is employed).

Automatic Water Level Controller

In the 'Automatic Water Level Controller For Sump/Submersible Pump'